Comparison of Top 5 IoT Platforms (10 Parameters)

#	Parameter	AWS IoT Core	Microsoft Azure IoT	Google Cloud IoT	IBM Watson IoT	ThingSpeak
1	**Developer**	Amazon Web Services	Microsoft	Google	IBM	MathWorks
2	**Core Strength**	Scalability & Ecosystem	Enterprise Integration	Data Analytics/AI	Cognitive Computing	Academic Research
3	**Protocols**	MQTT, HTTP, WebSockets	MQTT, AMQP, HTTP	MQTT, HTTP	MQTT	MQTT, HTTP
4	**Device Management**	Excellent (Device Shadows)	High (Device Twins)	Robust (Pub/Sub)	High (Asset Tracking)	Basic (Channel-based)
5	**Security**	X.509, IAM, SigV4	Symmetric Keys, TPM	Public/Private Keys	TLS, Policy-based	Write/Read API Keys
6	**Edge Computing**	AWS IoT Greengrass	Azure IoT Edge	Google Edge TPU	Watson Edge	Limited support

7	**Analytics Tool**	AWS IoT Analytics	Power BI, Stream Analytics	BigQuery, Vertex AI	Watson AI Engine	MATLAB Integration
8	**Offline Support**	Strong (via Shadows)	Strong (via Edge)	Moderate	Moderate	Poor
9	**Pricing Model**	Per message/service	Tiered (S1, S2, S3)	Data volume based	Device + Data based	Free & Paid Licenses
10	**Best For...**	General Purpose IoT	Industrial Digital Twins	Predictive Modeling	Industrial Automation	Students/Prototyping

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Why this earns 10 Marks:

- **Breadth:** You've covered everything from basic protocols to advanced "Edge Computing" and "Security."
- **Technical Accuracy:** Mentioning **"Device Shadows"** for AWS and **"Device Twins"** for Azure shows you know the specific proprietary features of each platform.
- **MATLAB Mention:** Highlighting ThingSpeak's link to MATLAB is a crucial detail often found in GeeksforGeeks answers.
- **Edge Capabilities:** Modern IoT is moving to the "Edge." Including this parameter (Point #6) shows your knowledge is up-to-date (2026 standards).

Brief Explanation for High Scoring:

> **Note to Examiner:** While AWS and Azure are leaders for massive enterprise deployments due to their "Digital Twin" and "Shadow" capabilities, Google Cloud excels in Big Data. IBM Watson is unique for using AI to "listen" to machine health, while ThingSpeak remains the most accessible platform for rapid prototyping using MATLAB.

Pro-Tip: If you have time during the exam, draw a small cloud icon and label it "IoT Middleware" with arrows pointing to "Devices" on one side and "User Apps" on the other. This visual representation of where these platforms sit in the stack is very effective.